Graph Theory – Introduction Notes

Discrete Mathematics — BCA Honours

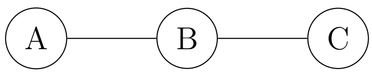
Based on Kenneth H. Rosen's *Discrete Mathematics and Its Applications with Combinatorics and Graph Theory*

1. Informal Introduction to Graphs

A graph is a pictorial representation of relationships between objects. It helps us model connections, such as:

- Friendships in social networks
- Routes between cities
- Flight connections
- Network topologies in computers

Example: Consider 3 cities: A, B, and C. There are roads between A–B and B–C. We can represent this as:



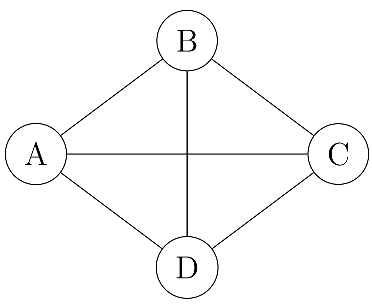
2. The Königsberg Bridge Problem

In the 18th century, the city of Königsberg (now Kaliningrad) had seven bridges connecting different parts of land separated by a river. People wondered:

"Is it possible to take a walk that crosses each bridge exactly once and return to the starting point?"

Euler's Insight: Represent the land areas as vertices and the bridges as edges, forming a graph. He showed that such a walk was not possible, laying the foundation of graph theory.

Graph Representation:



3. Formal Definitions (As per Rosen)

Graph:

A graph G is an ordered pair $G = (V, E)$, where V is a non-empty set of vertices and E is a set of edges (unordered or ordered pairs from V).

Finite and Infinite Graph:

A finite graph has a finite number of vertices and edges. An infinite graph has an infinite number of vertices and/or edges.

Simple Graph:

An undirected graph with no loops and no multiple edges between the same pair of vertices.

Multigraph:

An undirected graph that may have multiple edges between the same pair of vertices, but no loops.

Pseudograph:

A graph that may contain both multiple edges and loops.

Undirected Graph:

A graph where the edges are unordered pairs of vertices (i.e., edge $\{u, v\} = \{v, u\}$).

Directed Graph:

A graph where edges are ordered pairs (arcs), i.e., (u, v) is directed from u to v.

Mixed Graph:

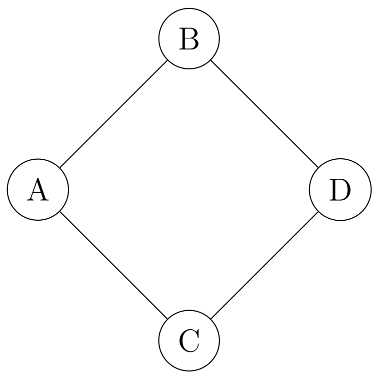
A graph that contains both directed and undirected edges.

4. Example

Problem: Draw a graph that models the following situation: Let A, B, C, D be people. A is friends with B and C. B is also friends with D. C is friends with D. Draw an undirected graph representing this.

Vertices: A, B, C, D

Edges: {A, B}, {A, C}, {B, D}, {C, D}



5. Recap Table

Term	Meaning
Graph	Set of vertices and edges
Simple Graph	No loops or multiple edges
Multigraph	Multiple edges, no loops
Pseudograph	Allows both multiple edges and loops
Directed Graph	Edges have direction (arrows)
Undirected Graph	Edges are bidirectional
Mixed Graph	Has both directed and undirected edges
Finite Graph	Limited vertices and edges
Infinite Graph	Unbounded vertices and/or edges

6. Exercise

Air Pollution Monitoring Network

Five cities have sensors installed that send data to each other for pollution tracking:

- Kochi shares data with Thrissur and Palakkad.
- Palakkad forwards data to Coimbatore.
- Coimbatore shares data with Madurai.
- Madurai sends data back to Kochi once daily.

Model the communication system as:

a) A directed graph showing flow of data.

b) A multigraph if data is sent multiple times per day.

c) A simple graph if we only care whether communication exists.

d) A pseudograph if a city processes and returns data to itself.